Econ 2 - Lecture 6 - 4/15/26

Lecture Quiz #3 Released Today, Due Monday, 4/20

Activity #2 this week

Coffee Hours tomorrow at 10 - 11AM

Midterm Exam on Monday, April 27th

Today: CPI & Unemployment

Next Week: Unemployment +

$$= \frac{\text{Nominal GDP} \rightarrow P_{CY} \times Q_{CY}}{\text{Real GDP} \rightarrow P_{BY} \times Q_{CY}} \times 100$$

Last Class: GDP → GDP Deflator → Inflation

GDP Inflation Measure includes all products in GDP

→ Excludes products not in GDP! Used cars

Consumer Price Index (CPI)

Only include products relevant to consumers

Step 1: Create a basket of goods & services

Include C, M (Consumption & Imports)

Exclude I, G, X (Investments, Gov't expenditures, Export)

Used goods included

Food, housing, coffee, shoes, cars,
Phones, TVs, etc.

Step 2: Weight each good in basket to represent how important the price is to the average consumer

Lasik   In 2000, Lasik Price = $10,000/eye
         In 2019, Lasik Price = $2,000/eye   80% drop

↓

0.3% weight

Food = 14%, Energy = 6%, Gas = 3%, Housing = 45%

Step 3: Calculate weighted basket cost → P × Q

Good	Base Year Weight/Q	Base Year P	Weighted Cost
Housing	30	100	30 × 100 = 3000
Food	50	10	50 × 10 = 500
Gas	20	4	20 × 4 = 80

$\underbrace{\text{Base Year}}_{\text{Set at 2016}}$ Basket Cost = 3000 + 500 + 80 = $3580

All of our price analysis (inflation) will use the same base year basket weights every year.

→ Hold quantities/weights constant

→ Evaluate new basket cost every year

In 2019:

BY = 2016

Good	Q_{BY}	P_{BY}	P_{2019}
Housing	30	100	120
Food	50	10	20
Gas	20	4	2

Weighted Basket Cost in 2019 (CY)

$$\text{Basket Cost}_{CY} = Q^H_{BY} \times P^H_{CY} + Q^F_{BY} \times P^F_{CY} + Q^G_{BY} \times P^G_{CY}$$

$$= 30 \times 120 + 50 \times 20 + 20 \times 2$$

$$= \$4640$$

CPI: Ratio of Basket Costs, relative to base year

→ Normalize the average price in base year to be 100

$$CPI_{CY} = \frac{\text{Weighted Basket Cost}_{CY}}{\text{Weighted Basket Cost}_{BY}} \times 100$$

$$= \frac{Q_{BY} \times P_{CY}}{Q_{BY} \times P_{BY}} \times 100$$

$$= \frac{4{,}640}{3{,}580} \times 100 = 129.60$$

Base Year CPI

BY = 2016

Evaluating CPI at CY → CY = 2016 → 1

$$CPI_{BY=CY=2016} = \frac{\cancel{Q_{BY} \times P_{CY=BY}}}{\cancel{Q_{BY} \times P_{BY}}} \times 100 = 100$$

Use Real Data on CPI

$CPI_{2020} = 258.1$, Friday, $CPI_{2026} = 330$

$$\text{Inflation Rate}_{20,26} = \frac{CPI_{26} - CPI_{20}}{CPI_{20}} \times 100$$

$$\text{Inf. Rate}_{20,26} = \frac{330 - 258.1}{258.1} \times 100 = 27.9\%$$

Stimulus/additional cash in 2020 - 2021
- ↳ No where to spend!
- ↳ Open up from pandemic → Spending ↑ → Prices ↑
- ↳ Prices are sticky → Don't change often (stuck)
  - ↳ Bigger price changes when they occur

Many potential shortcomings of CPI

CPI overstates inflation by 0.5 - 1.0% / year

↳ Important to avoid deflation (spiral)

Substitution Bias:

Detergent: Tide vs. Arm & Hammer

Today: Both are $10 / gallon

Next Period: Tide Price = $14, A&H = $10

Avg. Price = $12

More weight on A&H than Tide

Average Price paid by consumers < $12 average

Overstate detergent price

Outlet Bias:

Granola Bars @ Costco, 20 for $20 → $1 / bar

Albertsons → $2 / bar → $1.50 / bar

More weight on Costco purchases → avg price paid per bar < $1.50

Overstate price faced by consumers

Quality Bias

iPhone (SE) = $400 last year, new version = $450

Estimate how much better is new version

Quality-Adjust

storage, camera } Estimate $75 in value

Goal #1: High-standard of Living
↳ GDP → commonly used

Goal #2: Stable Prices → GDP Deflator & CPI

Goal #3: Full-Employment

What causes someone to become unemployed?

Categorize Reasons for Unemployment

1. Frictional Unemployment
Time lag in hiring process
To obtain a job: apply, interview, references, etc.

frictions

New grads → student working at Coral Tree
Stop working in June, apply for jobs,
don't get hired until August

Not necessarily indicative of a poor economy

2. Seasonal Unemployment
Predictive annual patterns in employment